

Technology Stack

Date	13 July 2026
Team ID	
Project Name	SkillEcho — Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. A well-chosen technology stack ensures the application is scalable, maintainable, and aligned with project requirements.

The table below details the specific technologies chosen for each component of the SkillEcho architecture along with the justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit 1.35.0	Rapid prototyping with a native audio-input widget, built-in session state, and zero frontend build tooling; supports free Community Cloud deployment.
2	Backend / Server-Side	FastAPI 0.111.0 + Uvicorn 0.29.0 (ASGI)	Async-capable REST API layer with automatic OpenAPI docs and Pydantic validation; Uvicorn provides a production-grade ASGI server with --reload support.
3	Database / Data Storage	SQLAlchemy 2.0.30 ORM → PostgreSQL (Neon) / SQLite + psycopg2-binary driver	SQLAlchemy models work identically against SQLite (dev) and PostgreSQL (prod) via a configurable DATABASE_URL; PostgreSQL enforces referential integrity via foreign keys; psycopg2-binary avoids system libpq dependencies.
4	Cloud / Hosting / Deployment	Streamlit Community Cloud (frontend) + Neon Serverless PostgreSQL (database)	Zero manual infrastructure provisioning; free-tier hosting for the dashboard; Neon offers zero-ops, serverless Postgres for persistent storage.
5	Version Control & CI/CD	Git / GitHub + run.sh startup script + venv311	Standard collaborative version control; run.sh unifies launching both the FastAPI backend (port 8000) and Streamlit frontend (port 8501) in one command; venv isolates dependencies.
6	AI / ML & Third-Party Libraries	OpenAI Whisper (tiny model)	State-of-the-art open-source ASR; the 39M-parameter tiny model fits the 1 GB RAM budget on Streamlit Cloud; fp16=False ensures CPU compatibility.
		Sentence-Transformers — Bi-Encoder (all-MiniLM-L6-v2) 2.7.0	Generates 384-dim L2-normalised embeddings; cosine similarity via dot product; lightweight enough for CPU inference of semantic similarity.
		Sentence-Transformers — Cross-Encoder (ms-marco-MiniLM-L-6-v2)	Jointly attends over both texts for a more precise semantic-accuracy score than the Bi-Encoder alone.
		Librosa + SoundFile + NumPy	De facto standard for audio DSP; extracts RMS energy, zero-crossing rate, and pause/silence ratio; SoundFile handles low-level audio I/O; NumPy underlies all numerical computation.

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		ReportLab 4.2.0 + Matplotlib + Pillow	Generates the in-memory PDF evaluation report (Platypus layout engine) including an embedded waveform chart rendered with Matplotlib and processed with Pillow.
		Requests 2.32.3 + python-multipart 0.0.9	Requests lets the Streamlit frontend call the FastAPI backend; python-multipart parses multipart form data for audio/PDF uploads (UploadFile, Form()).

Architecture Layer Mapping

Architecture Layer	Technologies	Responsibility
Presentation Layer	Streamlit 1.35.0	User interface: login forms, concept browser, audio recorder, scorecard dashboard, PDF download, history views.
Application Layer	FastAPI 0.111.0, Uvicorn 0.29.0	REST API routing, Pydantic request validation, pipeline orchestration, error handling.
AI / ML Layer	Whisper (tiny), Sentence-Transformers (MiniLM-L6-v2, ms-marco Cross-Encoder), Librosa	Speech-to-Text, Semantic Similarity, Cross-Encoder Accuracy, Acoustic Feature Extraction, Filler Word Analysis.
Business Logic Layer	scoring.py, reports.py	Deterministic weighted scoring matrix and PDF report generation.
Data Access Layer	SQLAlchemy 2.0.30, psycpg2-binary	ORM models, session management, schema migrations, seed data.
Persistence Layer	PostgreSQL (Neon), SQLite	Relational data storage across 10 normalised tables with cascading foreign keys.

System Requirements

Requirement	Specification
Python Version	3.11+
Operating System	Linux (Ubuntu 22.04+ recommended), macOS, Windows (WSL2)
Minimum RAM	1 GB (Streamlit Cloud constraint)
Disk Space	~500 MB for models + dependencies
Network	Internet required for Neon PostgreSQL; Whisper model download on first run
Ports	8000 (FastAPI), 8501 (Streamlit)